

PA 446 – Week 4 - Coding for Civic Data Applications
Fall 2026
Weekly Exercise 4

1. Create a new repository on GitHub and name it “*Weekly Exercise 4*”.
2. Clone the repository to bring all its associated files locally to your RStudio. (5pts)

[Moving forward, make sure to commit and push after performing each step.]

3. Create a new *WEEKLYEXERCISE.md* file and answer the following question. Once you have answered, commit and push to GitHub.
 - a. What is civic technology, and how can Smart Cities benefit from it? [Write at least 300 words and properly cite (using APA, MLA, or other citation style) the sources you use.] (15 points)
4. Create a new R Script “*WEEKLYEXERCISE4*” and check the name of the default branch. Report the name as a comment in the R Script. [Hint: install and load the relevant package] (6pts)
5. Create a new branch, and name it “*DEV*”. Make sure it appears on GitHub as well. (2pts)
6. Write the code to answer the following questions in the “*WEEKLYEXERCISE4*” R Script in the “*DEV*” branch. Use the data file *Olympics.csv* present in the repo. The data set contains information about participation in the Winter Olympic Games from 1980 to the present day. The set begins with the Games in 1980 because information on delegation sizes is readily available online during that time period. Geographic data come from the *CIA World Fact Book* and population and GDP figures are from the World Bank.

Variable	Description
ID	Unique number for each country in the data set
country	Name of country of participating Olympic delegation
year	Year that particular Olympic Games occurred
host	1 if host nation, 0 otherwise
athletes	Number of athletes in Olympic delegation
temp	Average high temperature (in Fahrenheit) in January if country is in Northern Hemisphere or July if country is in Southern Hemisphere (for largest city)
precipitation	Average precipitation (in mm) in January if country is in Northern Hemisphere or July if country is in Southern Hemisphere (for largest city)
elevation	Highest peak elevation within the country
gold	Total gold medals won
silver	Total silver medals won
bronze	Total bronze medals won
population	Country population (in 100,000)

GDP	Country gross domestic product
participate	Indicator of whether country participated in the Olympics that year (=1) or did not participate (=0)

[Again, make sure you commit and push after writing code for each question. Hint: Use packages *readr*, *tidyverse*, and *ggplot2* to answer the following.]

- a. Calculate a new variable, called *'total.medals'*, which is the sum of gold, silver, and bronze, and add it to the Olympic dataset. (2pts)
 - b. For each country, how many gold medals has it won? (2pts)
 - c. For each year, how many total medals were given out? (2pts)

7. Go to the main branch and use the same *"Olympics.csv"* dataset to answer the following questions. Write your code in the previously created *"WEEKLYEXERCISE4"* R Script.
 - a. Which countries had the largest delegation of athletes in 1992? Create a tibble that contains only the variables *country* and *athletes*. (2pts)
 - b. For the following five countries, plot the number of gold medals earned over time: United States, France, Germany, Russia, and China. (4pts)

8. Merge the *"DEV"* branch to the main branch. When resolving the merge conflicts, rather than selecting a part of the code, make sure to keep *all* the code. Once merged, the *"WEEKLYEXERCISE4"* R Script in the main branch should consist of all code answering Q4, Q6 and Q7. (10pts)

9. Create a new issue and name it *"WEEKLY EXERCISE 4"*. Label it as *"WEEKLY EXERCISE 4"* and place it under the milestone that should be named *"Weekly Exercises"*. Add a description to the issue and submit it. (10pts)

10. Share the GitHub repository link on the submission link on Canvas.